QUANTITATIVE MEASUREMENT · EXPERIMENTATION

Would they have bought anyway? Measuring true incrementality for Miss Selfridge

How a lifecycle display program was held to a causal standard — randomized holdouts, two-tailed significance testing, and a metric deliberately matched to the mission — delivering a 47% new customer rate that could actually be defended.

ROLE	TEAM	METHODS	OUTCOME
Measurement & Personalization Specialist	Capito · Qubit ecosystem · Miss Selfridge	Holdout design · Lifecycle segmentation · Significance testing	47% new customer rate, causally validated

OVERVIEW

Proving what marketing actually causes

The project centered on a question most marketing programs never honestly answer: of all the customers a campaign claims, how many would have bought anyway? For Miss Selfridge, a lifecycle display program had ambitious goals — find new customers, convert prospects, and grow lifetime value — against a relentless daily trade calendar. My focus was ensuring every result the program reported was a causal claim, not an attribution artifact.

Embedded in the Arcadia ecosystem, I worked on the measurement and audience architecture behind a creative-led, data-driven display strategy — one designed specifically to drive New Customer Rate rather than recycle existing demand.

01 — THE TENSION

Three ways the numbers could have lied

1 · THE ATTRIBUTION TRAP

Retargeting looks phenomenal on paper — it shows ads to people already deep in the funnel, then claims credit when they convert. Incremental ROI focused on retargeting alone is unnaturally inflated, because many of those customers would have purchased with no ad at all. The dashboard says success; the counterfactual says recycling.

2 · ONE BUDGET, THREE JOBS

Acquisition, prospecting-to-conversion, and lifetime-value growth compete for the same spend — and they don't share a success metric. Optimizing the blended number quietly starves the hardest job (finding genuinely new customers) to feed the easiest one (re-converting the loyal).

3 · A CALENDAR THAT NEVER SLEEPS

Miss Selfridge's trade plan changed daily — Black Friday, Cyber Monday, the Christmas run-up — with creative refreshed to match offline and online activity. Measurement had to survive constant creative churn: any framework that required a frozen campaign to produce a clean read was useless here.

> *The risk wasn't a failing campaign. It was a succeeding-looking campaign — one whose numbers could flatter without informing. The most dangerous dashboard is the one that's always green.*

02 — THE CRAFT

A measurement-first framework

Budget, audience, and metric were matched deliberately by funnel job — and everything was measured against a counterfactual. Explore the four intelligence layers below.

BUDGET FENCED BY FUNNEL JOB — NOT BY CHANNEL HABIT

50%	25%	25%
NEW CUSTOMERS	PROSPECT → CONVERSION	LIFETIME VALUE

Each job kept its own scorecard — no blended ROI reporting that lets retargeting subsidize acquisition's optics.

1 · Audience | 2 · Investment | 3 · Creative | **4 · Measurement**

PRINCIPLE

Every claim of impact must survive a comparison against "what if we'd done nothing?"

- 10% randomized holdout receiving the audience and investment logic — but served a blank ad.
- Incrementality = treated conversion minus holdout conversion.

03 — IMPLEMENTATION

Rigorous execution: the placebo arm

We delivered a **10% control group** — a true placebo arm: same audience intelligence, same investment logic, but a blank creative — so that the difference between arms could only be attributed to the advertising itself.

The lift was validated as statistically significant by a **two-tailed test** — two-tailed deliberately, because a campaign can plausibly hurt as well as help, and an honest test checks both directions. And we held a hard reporting line: **incremental ROI was never reported on its own** — always paired with New Customer Rate, so the program was judged on its mission, not on recycled demand.

MEASUREMENT ARCHITECTURE — RANDOMIZED HOLDOUT DESIGN

Treated arm — 90%
Audience intelligence + investment intelligence + **live creative**, refreshed daily against the trade plan

Holdout arm — 10%
Same audience + same investment logic, but served a **blank ad** — a true placebo

→

Incrementality
Treated conversion − holdout conversion, Validated by a two-tailed significance test.

MEASURED AGAINST THE COUNTERFACTUAL, NOT AGAINST ZERO

04 — THE EVIDENCE

Measuring the lift

47%
New Customer Rate — nearly half of campaign-driven purchases from customers entirely new to Miss Selfridge

10%
randomized placebo holdout — incremental lift confirmed by a two-tailed significance test

50/25/25
disciplined allocation — budget fenced by funnel job, protecting acquisition from retargeting's easy wins

Daily
creative-trade alignment — display refreshed against the trade calendar through peak season

> *"Our campaign was creative led, but data driven, and designed specifically to find completely new customers and increase existing customer lifetime value."*

THE GROWTH

Key learnings

ROI can lie
The single most important number in the program wasn't a lift — it was the holdout's conversion rate. Without the counterfactual, retargeting-heavy ROI flatters itself. With it, every claim became causal. This is the study that made "measured against the counterfactual, not against zero" my permanent measurement filter.

The metric must match the mission
A program whose mission is new customers cannot be judged on blended ROI. Elevating New Customer Rate to co-primary metric changed what the program optimized for — proof that metric selection is itself a design decision.

Two-tailed humility
Testing both directions is a statement of intellectual honesty: interventions can hurt. Building that assumption into the test — rather than only looking for wins — is what separates measurement from marketing.

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lifecycle display measurement within the Arcadia Group ecosystem. Proprietary metrics generalized for confidentiality.